Computer Science & IT

COMPILER DESIGN
(CD)

Parsers

Lecture No. 3



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Glammary

Topics to be Covered



- Topic
- Topic
- Topic

Grammar, parsey



Unstoppable Indians: Stories to Ignite Student Motivation

Ajay Piramal (b. 1955) transformed his family's modest pharmaceuticals business into the global Piramal Group. After acquiring Cadila Healthcare, he focused on contract research, strategic acquisitions, and digital healthcare platforms. His microfinance initiatives have empowered thousands of women entrepreneurs in rural India, while his philanthropic work in education and healthcare has improved access for underserved communities. Piramal's vision combines business acumen with social Impact. *Lesson:* Purpose-driven innovation and strategic diversification can drive growth and empower communities.



Unstoppable Indians: Stories to Ignite Student Motivation

Tulsi Tanti (b. 1958) founded Suzlon Energy in 1995, shifting from textiles to wind turbine manufacturing. Under his leadership, Suzlon became the world's third-largest wind turbine maker, deploying over 10 GW of capacity globally. Tanti championed local production, cost optimization, and rural job creation, navigating regulatory challenges and market swings. His work accelerated India's renewable energy sector and inspired cleantech entrepreneurs worldwide. ***Lesson:*** Visionary sustainability and adaptive execution can transform industries and uplift communities.



Unstoppable Indians: Stories to Ignite Student Motivation

Venu Srinivasan (b. 1951) modernized TVS Motor Company as its chairman, introducing new motorcycle models and expanding exports to over 60 countries. He fostered a culture of performance excellence and invested in electric mobility R&D.; Srinivasan also chairs Sundaram Clayton and leads CSR programs in education, healthcare, and skill development. His balanced approach to profitability and community engagement sets a benchmark in responsible industry leadership. ***Lesson:*** Combining operational excellence with community investment builds sustainable enterprises.



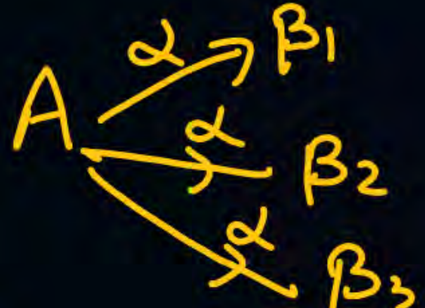
Unstoppable Indians: Stories to Ignite Student Motivation

Dr. Pankaj Patel (b. 1953) co-founded Zydus Cadila, focusing on affordable medicines and innovative vaccines. His leadership in developing India's first DNA vaccine and cost-effective treatments for chronic diseases has enhanced global health equity. Patel's philanthropic initiatives include healthcare outreach programs for rural populations. His commitment to high-quality R&D; and social responsibility highlights the role of science in serving humanity.

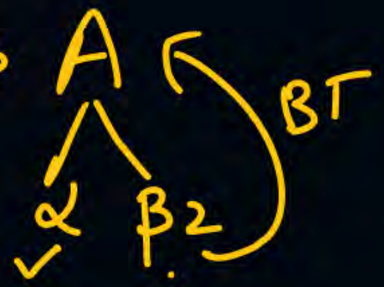
Lesson: Aligning innovation with accessibility ensures that breakthroughs benefit all sections of society.


G
am
unam
G
LR
RR
G
deter
non deter

Non deterministic grammars:-

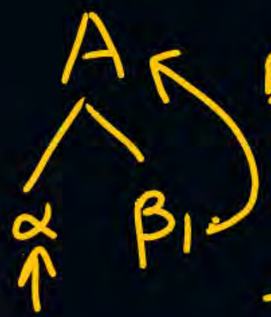


$A \rightarrow \alpha\beta_1 / \alpha\beta_2 / \alpha\beta_3$

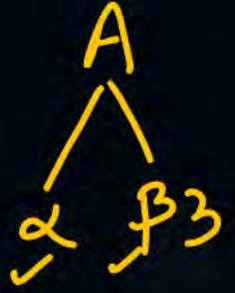


$\alpha\ \beta_3$

$A \rightarrow \alpha\beta_1$



Back tracking → going back



TDP cannot work with non deterministic grammars

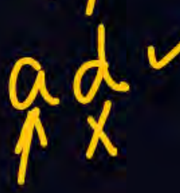
In non deterministic grammar, the problem is we are taking a decision without seeing enough of the string

$A \to \alpha\beta_1 / \alpha\beta_2 / \alpha\beta_3$



TDP cannot work with NDG.

$A \to aB / ac / ad$



$A \to \alpha A'$

$A' \to \beta_1 / \beta_2 / \beta_3$

} deterministic grammar



No back tracking in case of deterministic grammar

5 min

$S \to iEtS$
$\quad / \; iEtSeS$
$\quad / \; a$

$E \to b$

ambiguous

$iEtiEtseS$



(1) ✓

(2)



5 min

$S \to iEtS$

$/\ iEtSeS$

$/\ a$

$E \to b$

deterministic $\Longrightarrow$

$S \to iEtSS'/a$

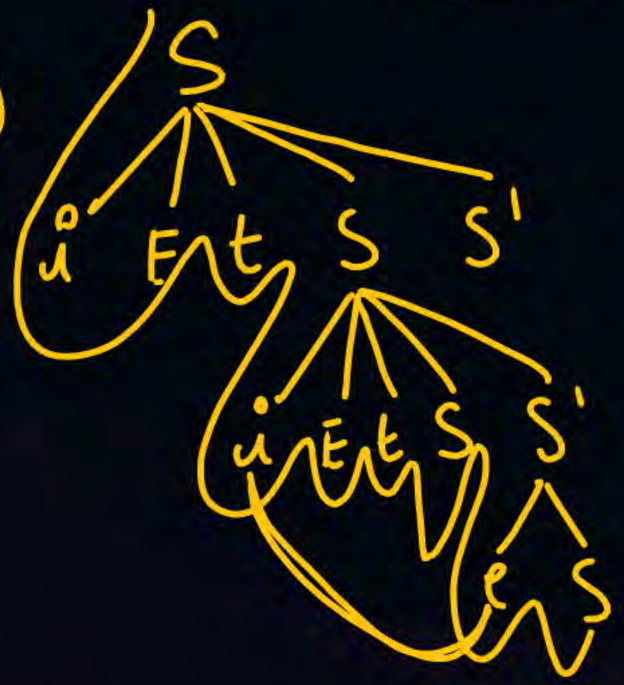
$S' \to \epsilon / eS$

$E \to b$

$\Rightarrow$ ambiguous

iEtiEtseS

me





Elimination of non determinism (left factactoring)

does not eliminate ambiguity

$S \rightarrow aSSbS$
$/aSaSb$
$/abb$
$/b$

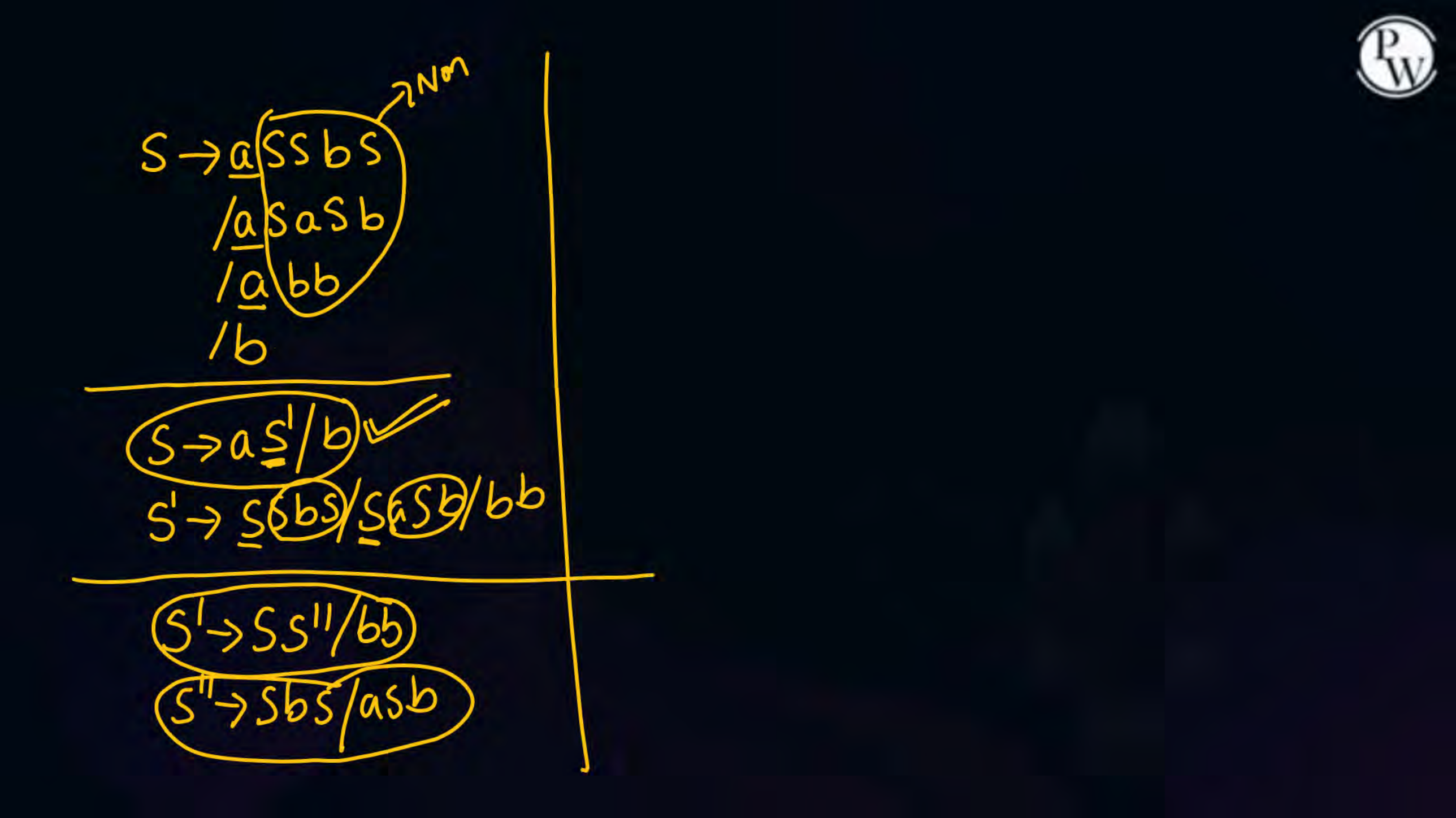
$S \rightarrow aS' / b$

$S' \rightarrow SbS / SaSb / bb$

$S' \rightarrow SS'' / bb$

$S'' \rightarrow SbS / aSb$

$S \rightarrow bSSaaS$
$/\ bSSaSb$
$/\ bSb$
$/\ a$


Non
S → a S S b S
/ a S a S b
/ a a b b
/ b
S → a S' / b
S' → S b S / S a S b / b b
S' → S S'' / b b
S'' → S b S / a S b

$S \rightarrow bSSaaS$

$/\ bSSaSb$

$/\ bSb$

$/\ a$

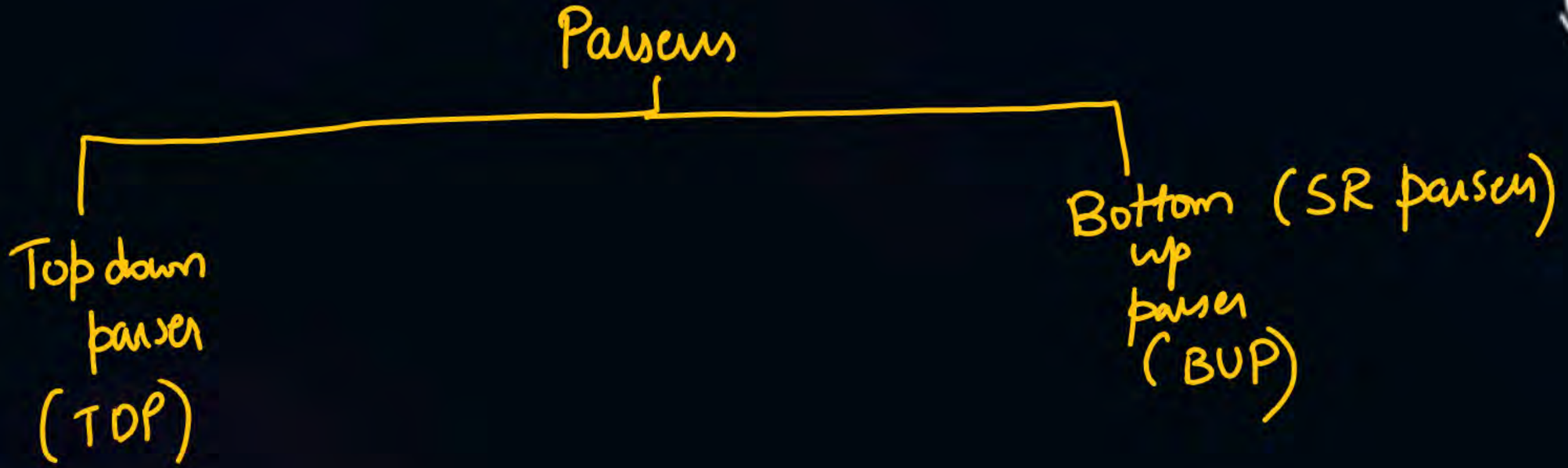
$S \rightarrow bSS' / a$

$S' \rightarrow SaaS / SaSb / b$

$S' \rightarrow SaS'' / b$

$S'' \rightarrow aS / Sb$

90% - passing :-


Parsers
Top down parser (TDP)
Bottom up parser (BUP)
(SR parsers)

Diff b/w TDP and BUP:-

$S \rightarrow aABe$

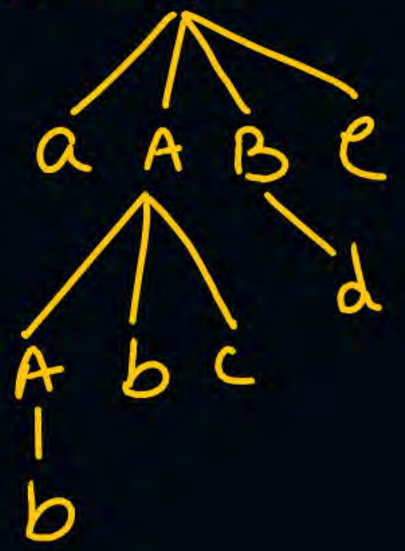
$A \rightarrow Abc/b$

$B \rightarrow d$

---

w: abbcde

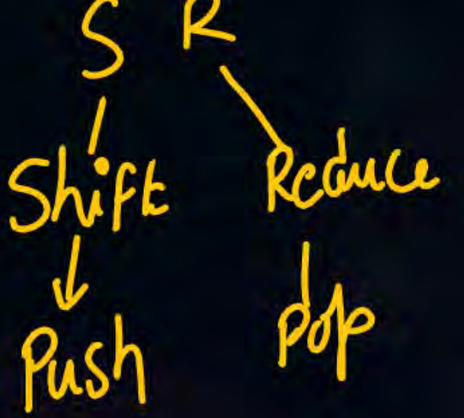
TDP :-



(SR)

BUP



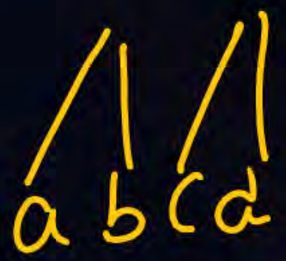


The main purpose of a parser is given a grammar and a string, we have to find out whether the string can be generated by grammar or not.

The main decision of TDP is what production to use.

$S \rightarrow aSb / bSa / aa$

The main decision of BUP is when to reduce



BU → LR parsers
- Left to right
- Reverse of RMD

$S \to aABe$
$A \to Abc \mid b$
$B \Rightarrow d$

---

abbcde

TDP uses LMD (left most derivation)

$S \to a\underline{A}Be$
$\to a\underline{A}bcBe$
$\to abbc\underline{B}e$
$\to abbcde$

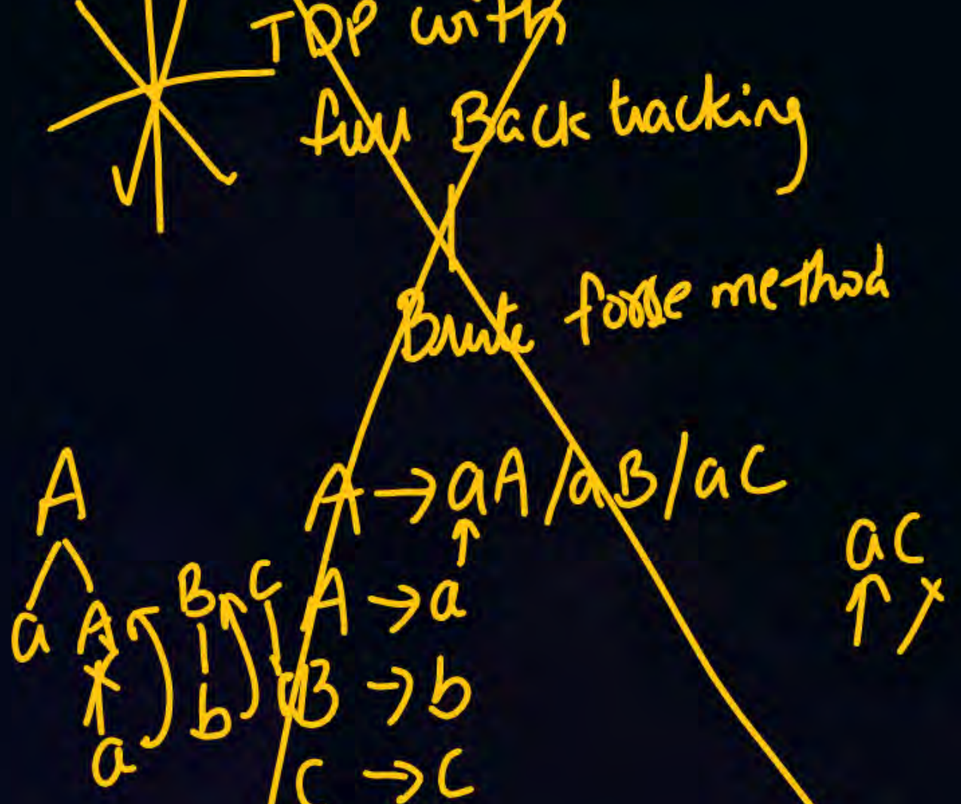
BUP uses Reverse of RMD

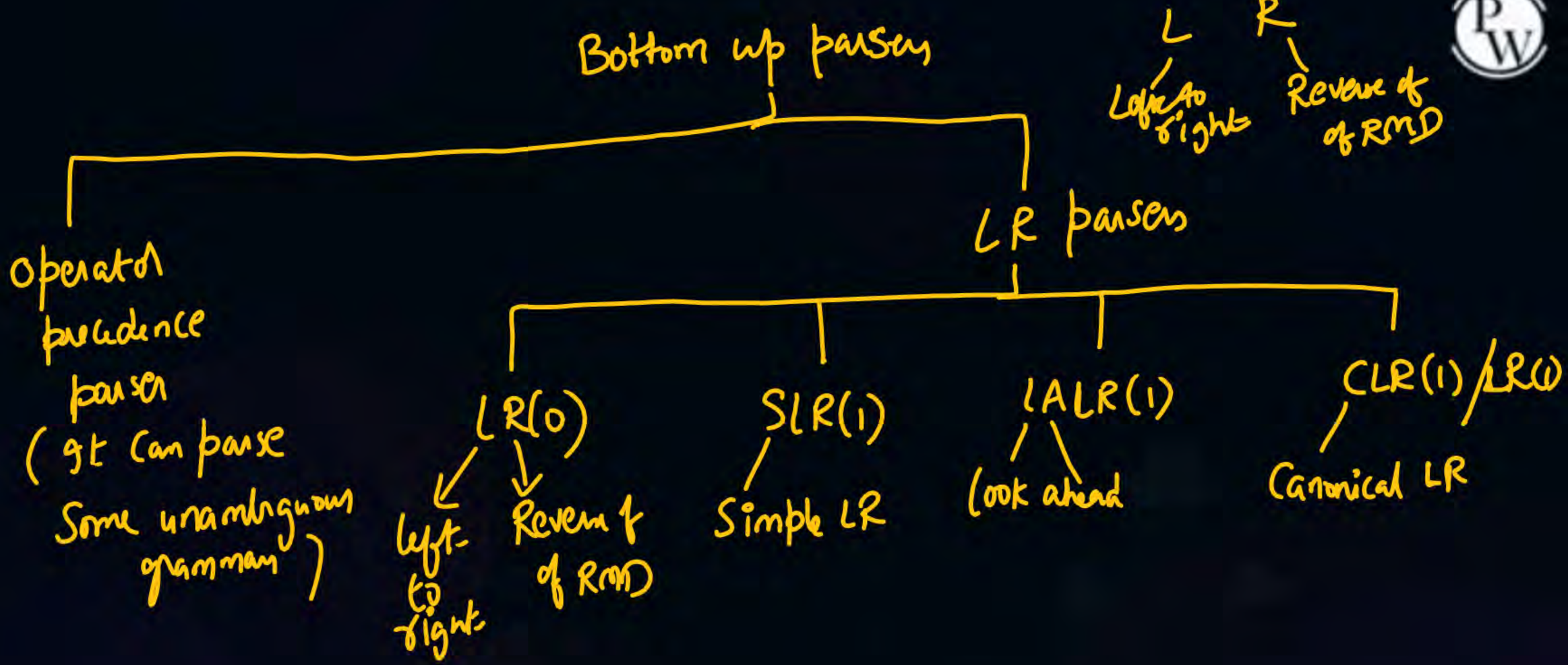
$S \Rightarrow aA\underline{B}e$
$\Rightarrow a\underline{A}de$
$\Rightarrow a\underline{A}bcde$
$\Rightarrow abbcde$

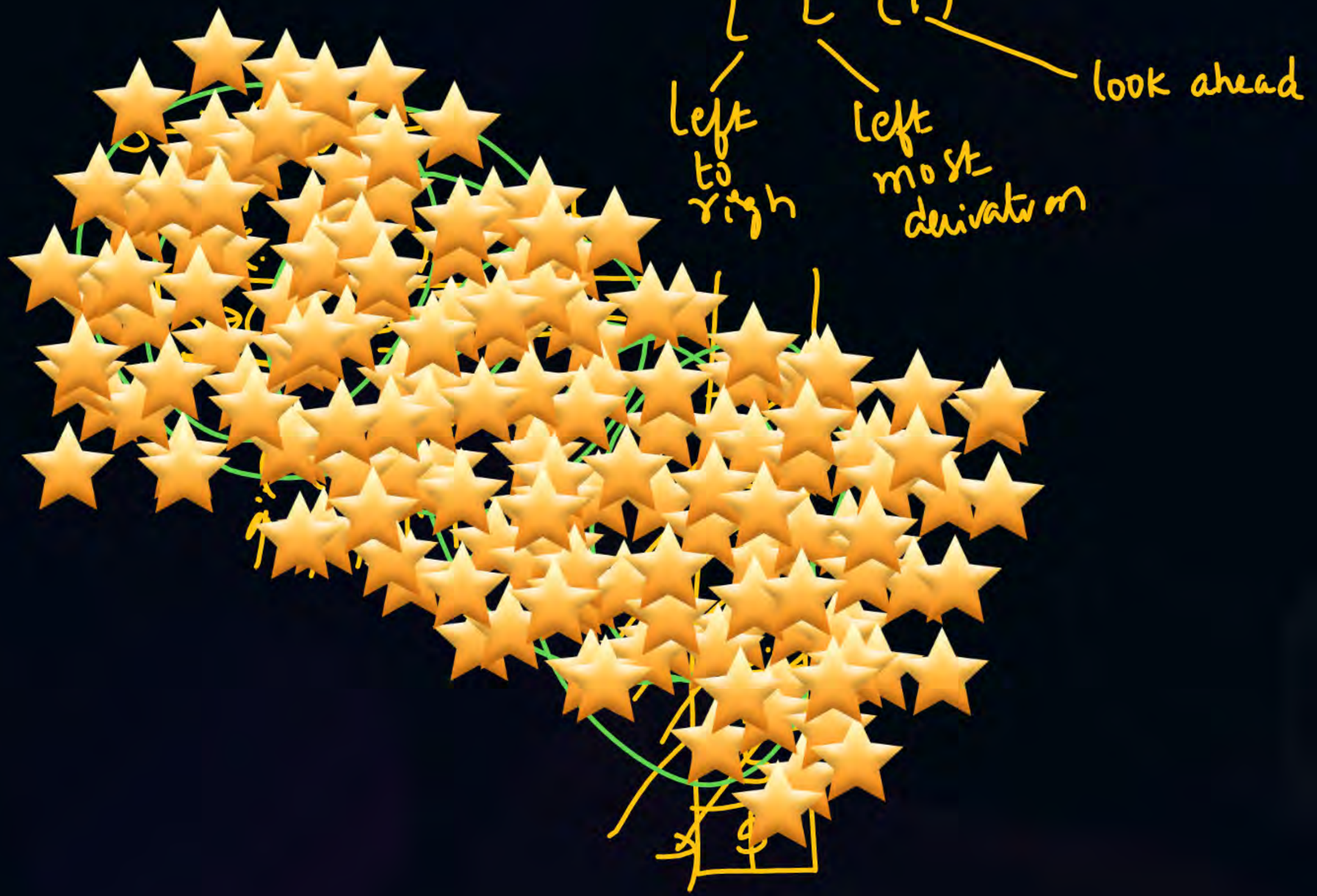


Top down parsers (TDP)

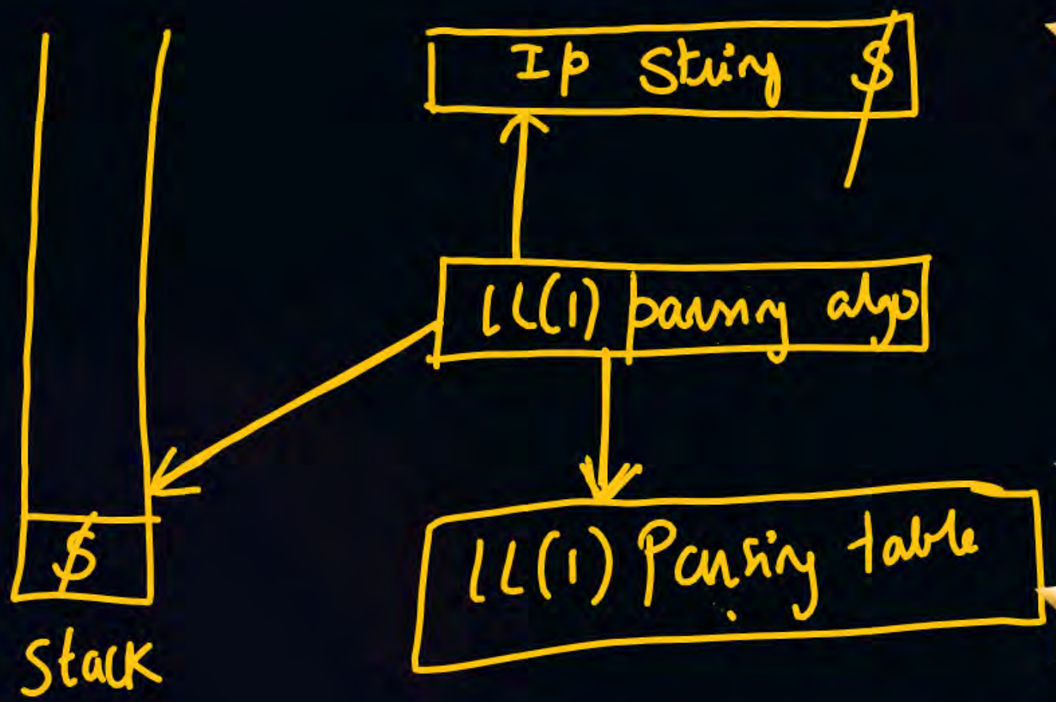
- TDP with full Backtracking
  - Brute force method
- TDP without Backtracking
  - ✗ Recursive descent
  - ✗ Non Recursive descent. LL(1)




Bottom up parsers
L
Left to right
R
Reverse of RMD
Operator precedence parser
(It can parse
Some unambiguous grammars)
LR parsers
LR(0)
Left to right
Reverse of RMD
SLR(1)
Simple LR
LALR(1)
Look ahead
CLR(1)/LR(1)
Canonical LR


LL(1)
left to right
left most derivation
look ahead

LL(1) parser





→ LL(1) parsing algorithm will take i/p from i/p buffer, top of the stack from stack and consults the LL(1) parsing table to take a decision

We need to first construct the parsing table using first and follow functions

First(α) gives the set of all terminals that begin the strings derived from α. If $\alpha \overset{*}{\Rightarrow} \in$ then $\in$ is also in first(α)

Rules to find first(α):-

→ If α is a terminal, then First(α) = α

Ex: First(a) = {a} first(id} = {id}

→ If α is a non terminal then there are two possibilities

a) $\alpha \to \epsilon$ then add 'ϵ' to First(α)
   $\alpha \overset{*}{\Rightarrow} \epsilon$ then add 'ϵ' to First(α)

b) $\alpha \to Y_1 Y_2 Y_3 \cdots Y_k$

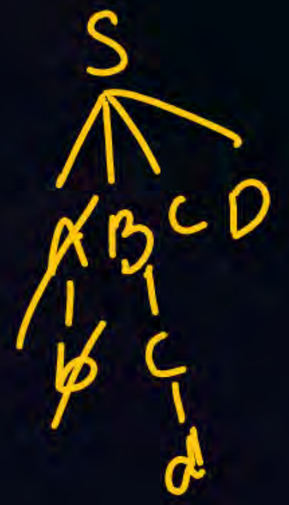
then First(α) = First(Y_1) if $Y_1 \not\Rightarrow \epsilon$

= First(Y_1) − {ϵ} ∪ First($Y_2 Y_3 \cdots Y_k$) if $Y_1 \overset{*}{\Rightarrow} \epsilon$

	First
$S \rightarrow aABCD$	$\{a\}$
$A \rightarrow b/\epsilon$	$\{b, \epsilon\}$
$B \rightarrow c/\epsilon$	$\{c, \epsilon\}$
$C \rightarrow d$	$\{d\}$
$D \rightarrow e$	$\{e\}$



	First
$S \to ABCD$	$\{b, d\}$
$A \to b/\epsilon$	$\{b, \epsilon\}$
$B \to c$	$\{d\}$
$C \to d$	$\{d\}$
$D \to e$	$\{e\}$



	First
$S \to ABCDE$	$\{a, b, c\}$
$A \to a/\epsilon$	$\{a, \epsilon\}$
$B \to b/\epsilon$	$\{b, \epsilon\}$
$C \to c$	$\{c\}$
$D \to d/\epsilon$	$\{d, \epsilon\}$
$E \to e/\epsilon$	$\{e, \epsilon\}$



	first
$S \to Bb/cd$	$\{a, b, c, d\}$
$B \to aB/\epsilon$	$\{a, \epsilon\}$
$C \to cC/\epsilon$	$\{c, \epsilon\}$

C c



11:15 PM

YT Gatewall Engle

2 hour → 4 marks

THANK - YOU